

CAPE REINGA, New Zealand

WMO: 930040

Lat: 34.4294S

Lon: 172.6817E

Elev: 697

StdP: 14.33

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	47.2	48.7	37.5	33.8	50.1	39.5	36.5	50.7	54.6	55.5	50.0	54.0	19.3	200	0.777

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.5	73.8	67.4	72.5	66.5	71.3	65.8	70.0	71.7	68.8	70.5	67.8	69.4	16.9	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.4	111.0	70.6	68.2	106.3	69.6	67.1	102.4	68.7	34.5	71.6	33.4	70.5	32.6	69.5	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
48.4	42.7	36.9		DB	44.6	76.9	1.6	1.5	43.5	78.0	42.5	78.9	41.6	79.7	40.5	80.8
				WB	40.6	72.1	1.5	1.1	39.5	72.9	38.6	73.6	37.7	74.2	36.6	75.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.2	65.8	67.2	65.8	63.1	59.8	56.3	54.8	54.5	55.9	57.2	59.4	63.0	(6)
(7)		DBStd	5.03	2.36	2.32	2.33	2.49	2.81	2.68	2.45	2.27	2.38	2.27	2.39	2.45	(7)
(8)		HDD50	4	0	0	0	0	0	1	1	2	0	0	0	0	(8)
(9)		HDD65	1925	19	5	19	66	163	260	316	325	274	241	167	70	(9)
(10)		CDD50	3728	491	483	490	393	302	190	150	142	176	224	283	404	(10)
(11)		CDD65	176	45	68	44	9	1	0	0	0	0	0	0	9	(11)
(12)		CDH74	39	9	25	5	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	18.8	17.8	17.1	17.6	17.8	19.8	20.4	20.5	19.7	19.2	19.6	18.6	17.7	(14)
(15)	Precipitation	PrecAvg	47.1	2.3	3.3	2.8	3.7	4.6	5.2	5.8	4.9	3.9	3.1	2.7	3.7	(15)
(16)		PrecMax	73.2	6.5	11.1	7.2	12.1	8.7	14.1	12.5	8.5	8.1	6.1	5.9	8.3	(16)
(17)		PrecMin	24.7	0.5	0.7	0.4	0.6	1.2	2.2	1.8	2.3	2.0	0.8	0.8	0.5	(17)
(18)		PrecStd	11.6	1.5	2.8	2.0	2.7	1.9	2.3	2.6	1.8	1.8	1.5	1.3	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.3	76.7	74.5	71.5	68.1	63.8	62.2	61.8	63.9	66.3	68.5	72.4	(19)
(20)		MCWB	68.0	69.9	67.3	65.3	63.5	61.4	59.3	57.2	58.8	60.6	62.5	66.2	(20)	
(21)		2%	DB	73.1	74.5	72.8	69.6	66.2	62.3	60.5	60.3	62.1	64.2	66.6	70.4	(21)
(22)		MCWB	66.7	68.3	66.3	64.8	62.8	60.2	58.0	56.8	57.8	59.3	61.5	64.6	(22)	
(23)		5%	DB	71.7	73.0	71.4	68.2	64.7	61.2	59.3	59.1	60.8	62.5	65.1	68.8	(23)
(24)		MCWB	66.0	67.1	65.7	64.2	61.9	59.0	56.7	56.0	57.2	58.3	60.5	63.9	(24)	
(25)		10%	DB	70.4	71.5	69.9	67.0	63.6	60.1	58.3	58.0	59.5	61.1	63.7	67.3	(25)
(26)		MCWB	65.4	66.4	65.0	63.4	61.0	57.9	55.7	55.2	56.6	57.7	59.8	63.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.4	72.0	70.3	68.9	65.9	63.0	61.1	60.1	60.9	62.3	65.4	68.4	(27)
(28)		MCDWB	72.3	73.7	71.7	69.4	66.4	63.2	61.4	60.4	62.1	64.4	66.4	69.8	(28)	
(29)		2%	WB	69.0	70.7	68.8	67.3	64.3	61.3	59.2	58.4	59.4	60.9	63.3	66.8	(29)
(30)		MCDWB	71.0	72.6	70.4	68.1	65.2	61.7	59.7	59.1	60.5	62.3	64.6	68.1	(30)	
(31)		5%	WB	67.8	69.4	67.6	65.9	63.1	60.1	57.8	57.3	58.6	59.9	62.2	65.7	(31)
(32)		MCDWB	69.8	71.2	69.5	67.0	64.1	60.7	58.6	58.1	59.7	61.3	63.6	67.0	(32)	
(33)		10%	WB	66.8	68.2	66.5	64.6	61.9	58.8	56.7	56.2	57.6	58.9	61.2	64.7	(33)
(34)		MCDWB	68.9	69.9	68.6	65.9	63.1	59.7	57.7	57.3	58.9	60.4	62.8	66.0	(34)	
(35)	Mean Daily Temperature Range		MDBR	8.6	8.5	8.2	7.1	6.1	5.6	5.8	6.3	7.0	7.6	8.0	8.1	(35)
(36)		5% DB	MCDBR	10.0	10.3	9.9	8.5	6.9	6.0	6.5	7.2	8.0	9.2	9.4	9.8	(36)
(37)		MCWBR	5.7	5.8	5.8	5.7	5.2	5.5	5.6	5.7	5.9	6.3	6.4	6.2	(37)	
(38)		5% WB	MCDBR	7.8	7.7	7.7	6.2	5.7	4.9	5.3	5.5	6.6	7.4	7.5	7.2	(38)
(39)		MCWBR	5.4	5.4	5.4	5.1	5.1	5.5	5.5	5.3	5.5	6.0	6.1	5.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.336	0.335	0.323	0.313	0.304	0.304	0.300	0.308	0.326	0.326	0.326	0.332	(40)	
(41)		taud	2.621	2.654	2.679	2.699	2.693	2.662	2.680	2.647	2.572	2.587	2.614	2.647	(41)	
(42)		Ebn at Noon	318	312	304	290	276	267	275	287	298	311	318	320	(42)	
(43)		Edh at Noon	32	30	27	24	21	20	21	24	30	31	32	31	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2256	2014	1640	1216	876	718	776	1058	1437	1840	2095	2169	(44)	
(45)		RadStd	130	145	90	72	42	40	44	55	78	100	121	173	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (2 neighbors)	(m) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page